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🏢 WORK EXPERIENCE

- **Jul. 2023 – present, Xi'an University of Architecture and Technology, Xi'an, China**

Faculty-oriented postdoctoral fellow, School of Civil Engineering

🎓 EDUCATION

- **Sep. 2016 – Feb. 2023, Tongji University, Shanghai, China**

Ph.D. of Engineering, Department of Structural Engineering, College of Civil Engineering

Supervisor: Prof. Jian-Bing Chen

Dissertation: Seismic-Reliability-Based Structural Optimization by Probability Density Evolution Method

- **Jul. 2021 – Jul. 2022, Leibniz Universität Hannover, Hannover, Germany**

Visiting Ph.D. Student, Institute for Risk and Reliability

Co-Supervisor: Prof. Michael Beer

- **Sep. 2012 – Jun. 2016, Harbin Institute of Technology, Harbin, China**

Bachelor of Engineering, College of Civil Engineering

📊 RESEARCH INTERESTS

1. Structural optimization
2. Computational stochastic dynamics
3. Structural reliability analysis
4. Reliability-based design optimization
5. Reliability-based topology optimization

📖 PUBLICATIONS

📄 Journal Papers

1. Wan ZQ, Xie YQ, **Yang JS***, Lyu MZ, Ghafoori E. Accelerated stochastic model updating via decoupled multi-probability density evolution method with augmented change of probability measure. *Mechanical Systems and Signal Processing* 246: 113896. (IF=8.9, JCR Q1)
2. Xiao R, Sushama L, Meguid M, Zayed T, **Yang JS***. (2026) Time-variant reliability analysis of corroded gas pipelines considering dynamic defect evolution and interaction. *Reliability Engineering & System Safety* 267: 111899. (IF=11.0, JCR Q1)
3. Chen JB*, Weng LL, **Yang JS**. (2026) Dynamic-reliability-based design optimization by quantum-inspired algorithm and probability density evolution method combined with change of probability measure. *Journal of Vibration Engineering* 39(1):239-248. (In Chinese, EI)

4. **Yang JS**, Lyu MZ*, Chen JB, Xue JY. (2025) Structural design optimization under stochastic excitations considering first-passage probability constraint based on dimension-reduced probability density evolution equation. *Reliability Engineering & System Safety* 264: 111378. (IF=11.0, JCR Q1)
5. **Yang JS**, Wan ZQ*, Jensen HA. (2025) An efficient strategy for information reuse in probability density evolution method considering large shift of distributions with multiple random variables. *Probabilistic Engineering Mechanics* 79: 103728. (IF=3.0, JCR Q1)
6. Lyu MZ, **Yang JS***, Chen JB, Li J. (2025) High-efficient non-iterative reliability-based design optimization based on the design space virtually conditionalized reliability evaluation method. *Reliability Engineering & System Safety* 254: 110646. (IF=9.4, JCR Q1)
7. Chen G, Liu Y, Ma ZF, **Yang JS**, Beer M, Chian SC*, Wei SJ*. (2025) Assessing extent of building damage following an earthquake: Case study of the 2023 Turkey-Syria doublet. *npj Natural Hazards* 2: 51.
8. **Yang JS**, Chen JB*, Beer M. (2024) Seismic topology optimization considering first-passage probability by incorporating probability density evolution method and bi-directional evolutionary structural optimization. *Engineering Structures* 314: 118382. (IF=5.5, JCR Q1)
9. Weng LL, Acevedo CH, **Yang JS**, Valdebenito MA , Faes MGR , Chen JB*. (2024) An approximate decoupled reliability-based design optimization method for efficient design exploration of linear structures under random loads. *Computer Methods in Applied Mechanics and Engineering* 432: 117312. (IF=6.9, JCR Q1)
10. Chen G*, **Yang JS**, Wang RH, Li KQ, Liu Y, Beer M. (2024) Response to discussion of “Seismic damage analysis due to near-fault multipulse ground motion” . *Earthquake Engineering & Structural Dynamics* 53(2): 861–866. (IF=4.5, JCR Q1)
11. Chen G*, **Yang JS**, Liu Y, Kitahara T, Beer M. (2023) An energy-frequency parameter for earthquake ground motion intensity measure. *Earthquake Engineering & Structural Dynamics* 52(2): 271-284. (IF=4.060, JCR Q1)
12. Weng LL, **Yang JS**, Chen JB*, Beer M. (2023) Structural design optimization under dynamic reliability constraints based on probability density evolution method and quantum-inspired optimization algorithm. *Probabilistic Engineering Mechanics* 74: 103494 (IF=2.6, JCR Q1)
13. Chen G*, **Yang JS**, Wang RH, Li KQ, Liu Y, Beer M. (2023) Seismic damage analysis subjected to near-fault multi-pulse ground motion. *Earthquake Engineering & Structural Dynamics* 52(15): 5099-5116. (IF=4.5, JCR Q1, **Top cited and top viewed paper of the journal EESD**)
14. Feng CX*, Faes MGR, Broggi M, Dang C, **Yang JS**, Zheng ZB, Beer M, (2023). Application of interval field method to the stability analysis of slopes in presence of uncertainties. *Computers and Geotechnics* 153: 105060. (IF=5.218, JCR Q1)
15. **Yang JS**, Chen JB*, Beer M, Jensen HA. (2022) An efficient approach for dynamic-reliability-based topology optimization of braced frame structures with probability density evolution method. *Advances in Engineering Software* 173: 103196. (IF=4.255, JCR Q1)
16. **Yang JS**, Jensen HA, Chen JB*. (2022) Structural optimization under dynamic reliability constraints utilizing probability density evolution method and metamodels in augmented input space. *Structural and Multidisciplinary Optimization* 65(4): 107. (IF=4.542, JCR Q1)
17. **Yang JS**, Chen JB*, Jensen HA. (2022) Structural design optimization under dynamic reliability constraints based on the probability density evolution method and highly-efficient sensitivity analysis. *Probabilistic Engineering Mechanics* 68: 103205. (IF=3.350, JCR Q1)
18. **Yang JS**, Chen JB*. (2022) Structural optimization under dynamic reliability constraints by synthesizing change of probability measure and globally convergent method of moving asymptotes. *Journal of Vibration Engineering* 35(1):72-81. (In Chinese, EI, **High-influential paper of CNKI**)
19. Chen JB*, **Yang JS**, Jensen HA. (2020) Structural optimization considering dynamic reliability constraints

via probability density evolution method and change of probability measure. *Structural and Multidisciplinary Optimization* 62(5): 2499-2516. (IF=3.377, JCR Q1)

Conference Papers & Presentations

1. **Yang JS**, Lyu MZ, Chen JB. (2024) Dynamic-reliability-based optimization of high-dimensional nonlinear structures under stochastic excitations using dimension-reduced probability density evolution equation. The 10th International Workshop on Reliable Engineering Computing (REC 2024). Beijing, China.
2. **Yang JS**, Ding YQ. (2024) Seismic-reliability-based topology optimization of frame structures considering stochastic ground motions. The 9th International Symposium on Reliability Engineering and Risk Management (ISRERM 2024). Hefei, China. (**Invited lecture of MS14**)
3. **Yang JS**, Wan ZQ. (2024) Dynamic-reliability-based design optimization via probability density evolution method and efficient information reuse strategy. The 4th International Conference on Vulnerability and Risk Analysis and Management (ICVRAM2024) & The 8th International Symposium on Uncertainty Modelling and Analysis (ISUMA2024). Shanghai, China. (**Invited lecture of MS06**)
4. **Yang JS**, Chen JB. (2024) Reliability-based topology optimization of frame structures under seismic excitations by probability density evolution method. The 4th International Conference on Vulnerability and Risk Analysis and Management (ICVRAM2024) & The 8th International Symposium on Uncertainty Modelling and Analysis (ISUMA2024). Shanghai, China.
5. Weng LL, **Yang JS**, Chen JB. (2023) Dynamic reliability-based design optimization using quantum-inspired algorithm and probability density evolution method assisted by change of probability measure strategy. The 14th International Conference on Applications of Statistics and Probability in Civil Engineering (ICASP14). Dublin, Ireland. (Full paper)
6. **Yang JS**, Chen JB, Jensen HA. (2022) A gradient-free approach for dynamic-reliability-based design optimization based on the probability density evolution method. The 13th International Conference on Structural Safety & Reliability (ICOSSAR2021-2022). Shanghai, China. (Full paper)
7. **Yang JS**, Chen JB, Jensen HA. (2021) An efficient approach for dynamic-reliability-based design optimization using the probability density evolution method. The 14th World Congress on Structural and Multidisciplinary Optimization (WCSMO14). Boulder, USA.
8. **Yang JS**, Chen JB, Jensen HA. (2020) A PDEM-based method for structural optimization under dynamic reliability constraints. The 7th International Symposium on Reliability Engineering & Risk Management (ISRERM2020). Beijing, China. (Full paper, **Student scholarship award**)
9. **Yang JS**, Chen JB, Jensen HA. (2020) An efficient method for sensitivity estimation of failure probability based on the probability density evolution method. The 24th Annual Conference of HKSTAM 2020 & The 16th Shanghai –Hong Kong Forum on Mechanics and Its Application. Hong Kong, China. (**Best student presentation award**)
10. **Yang JS**, Chen JB, Jensen HA, Li J. (2019) Structural optimization considering dynamic reliability constraints via probability density evolution method. Proceedings of the 13th World Congress on Structural and Multidisciplinary Optimization (WCSMO13), 128-133. Beijing, China. (Full paper)
11. **Yang JS**, Chen JB, Jensen HA. (2019) An integrated method based on the probability density evolution method for dynamic reliability-based design optimization. The 32th KKHTCNN Symposium on Civil Engineering. Daejeon, South Korea. (Full paper)

Code, Reports & Preprints

1. Chen JB, Lyu MZ, **Yang JS**, Sun TT, Luo Y. (2023) MATLAB and Julia toolboxes for physically-driven DR-PDEE. (Code repository: <https://github.com/JCERSM/DR-PDEE-MATLAB> & <https://github.com/JCERSM/DR-PDEE-Julia>)

2. **Yang JS**, Chen JB. (2019) MATLAB toolbox for Probability Density Evolution Method (PDEM). (Code repository: https://github.com/Tree-Yang/Process_Oriented_PDEM & <https://github.com/Tree-Yang/GF-discrepancy-based-point-selection>)

PROJECTS

- 2025.01-2027.12, Young Scientists Fund (No. 52408212), National Natural Science Foundation of China (NSFC).
- 2024.01-2026.07, Special grants (No. 2023BSHTBZZ39), Postdoctoral Research Fund of Shaanxi Province, China.
- 2023.07-2026.07, Start-up funding for introduced talent (No. 1608724027), Xi'an University of Architecture and Technology, China.

SCHOLARSHIPS & AWARDS

- 2025, Top cited and top viewed article in Earthquake Engineering & Structural Dynamics, Wiley.
- 2023, High-influential research paper, CNKI.
- 2021, Scholarship for studying abroad (12 months), Chinese Scholarship Council (CSC).
- 2020, Student scholarship award, the 7th International Symposium on Reliability Engineering and Risk Management (ISRERM2020).
- 2020, Best student presentation award, the 24th Annual Conference of Hong Kong Society of Theoretical and Applied Mechanics.
- 2020, First prize scholarship for civil engineering, Tongji University.
- 2016, Excellent doctoral (freshman) scholarship, Tongji University.
- 2015, First prize scholarship, Harbin Institute of Technology.
- 2013, National scholarship of China.
- 2013, Merit student, Harbin Institute of Technology.

ACADEMIC SERVICE

- Member of Specialized Committee on Safety of Urban Lifeline Engineering, China Association for Engineering Construction Standardization (CECS), 2023-
- Reviewer of international journals including:
 - (1) Structural and Multidisciplinary Optimization
 - (2) Computer Methods in Applied Mechanics and Engineering
 - (3) Reliability Engineering & System Safety
 - (4) Mechanical Systems and Signal Processing
 - (5) Engineering Structures
 - (6) Probabilistic Engineering Mechanics
 - (7) Chinese Journal of Aeronautics
 - (8) ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part A: Civil Engineering
 - (9) Journal of Building Engineering
 - (10) Results in Engineering
 - (11) World Earthquake Engineering (In Chinese)
- Member of International Society for Structural and Multidisciplinary Optimization (ISSMO).